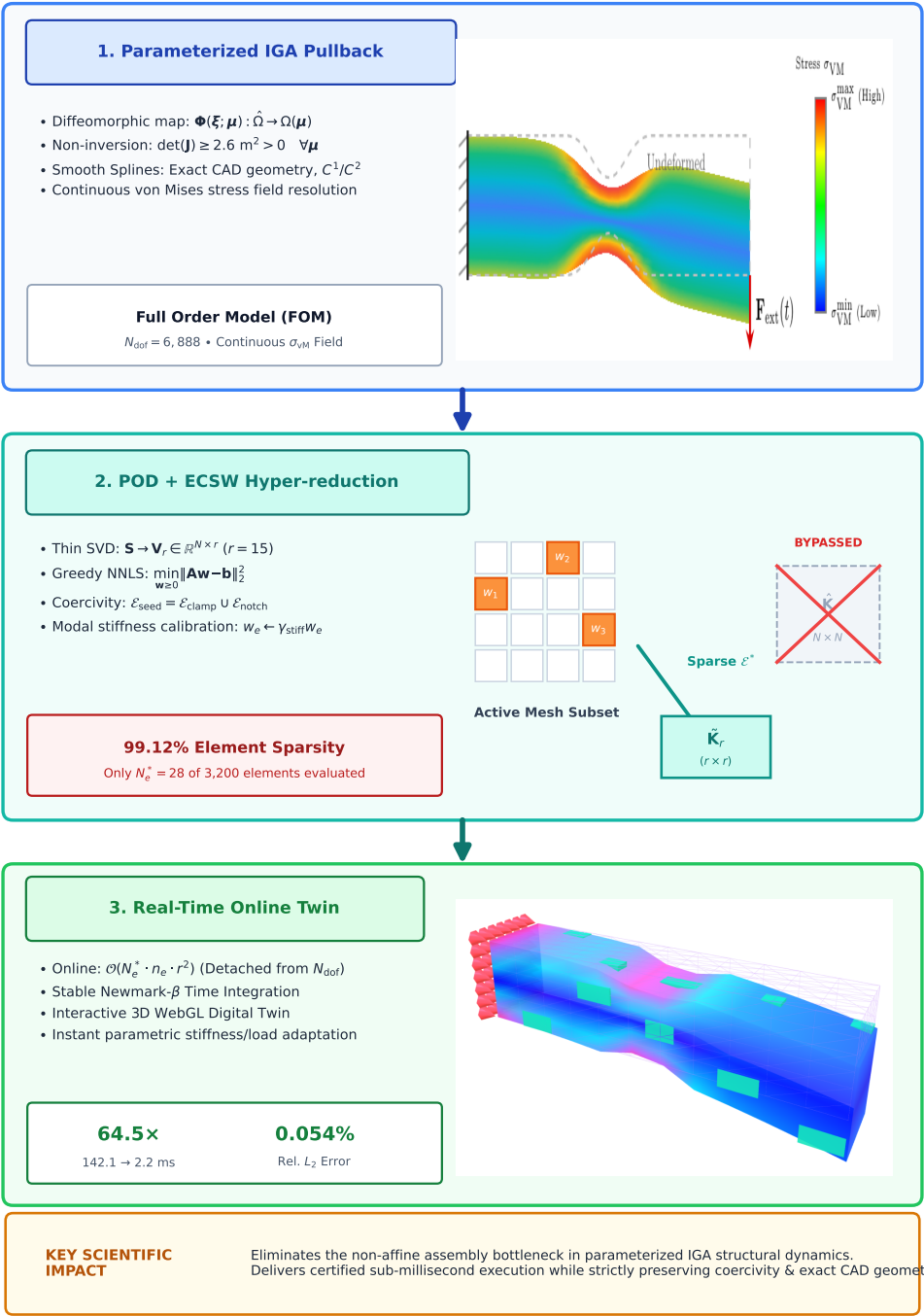


Graphical Abstract

Projection-based Reduced Order Modeling for Structural Dynamics via Isogeometric Analysis and Hyper-Reduction

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Highlights

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- Parameterized structural dynamics formulation based on Isogeometric Analysis.
- Geometry-preserving mapping onto an invariant reference domain for shape changes.
- Projection-based Reduced Order Modeling (pROM) via Proper Orthogonal Decomposition.
- Structure-preserving ECSW hyper-reduction for fast non-affine evaluations.
- Substantial speedups (64.5x in 2D, 96.9x in 3D) with relative errors below 0.06%.

Projection-based Reduced Order Modeling for Structural Dynamics via Isogeometric Analysis and Hyper-Reduction

Mumen Wehbe^{a,*}

^aConservatoire National des Arts et Métiers (Cnam), 292 Rue Saint-Martin, 75003 Paris, France

Abstract

Parameterized structural dynamics often demands repeated solution of the equations of motion across varying geometric configurations and operating conditions. Traditional numerical approaches rely on Finite Element methods where shape variations necessitate frequent remeshing and projection across incompatible meshes. This paper presents an integrated framework combining Isogeometric Analysis (IGA) with Projection-based Reduced Order Modeling (pROM) and hyper-reduction for structural dynamics. By exploiting the exact geometry description offered by B-splines and Non-Uniform Rational B-Splines (NURBS), parameter-dependent physical geometries are mapped back to a parameter-independent reference parametric domain. This eliminates the requirement for spatial remeshing and ensures consistent algebraic degrees of freedom across the parameter space. Proper Orthogonal Decomposition (POD) and Galerkin projection yield a drastically reduced-order system of dynamic equations. To address the computational bottleneck introduced by non-affine parameter-dependent metric tensors and geometric mappings, hyper-reduction is applied to decouple online evaluation complexity from the full-order mesh dimension. Numerical verification on representative 2D and 3D transient elastodynamic benchmarks demonstrates substantial dimensionality reduction, computational acceleration, and high accuracy.

Keywords: Isogeometric Analysis, Projection-based Reduced Order Modeling, Proper Orthogonal Decomposition, Structural Dynamics, Reference Domain Mapping, Hyper-Reduction

Nomenclature

Latin Symbols

$u, \dot{u}, \ddot{u}$	Displacement, velocity, and acceleration fields
$\hat{u}, q_r$	Full control variable and reduced coordinate vectors
M, C, K	Full-order mass, damping, and stiffness matrices
M_r, C_r, K_r	Reduced mass, damping, and stiffness matrices
V_r, P_A	POD projection basis and parameterized control points
F_{ext}, F_r	Full and reduced external dynamic load vectors
S, U, Σ, W	Snapshot and SVD factor matrices
σ_i, ϕ_i, ψ_i	Singular values and left/right singular vectors
$N_{i,p}, R_i^p$	B-spline and NURBS basis functions
N_{dof}, r	Full degrees of freedom and reduced basis rank
N_e, N_e^*	Total mesh elements and ECSW active element subset
$w_e, \gamma_{\text{stiff}}$	ECSW integration weights and stiffness scale factor

Greek Symbols

$\Omega(\mu), \hat{\Omega}$	Parameterized physical and invariant reference domains
$\hat{\Omega}_{\text{param}}, \xi, \mu$	Parametric unit domain, parametric coordinates, and parameter vector
Ξ, ε, σ	Knot vector, strain, and Cauchy stress tensors
$\rho, \mathbb{C}$	Mass density and fourth-order elasticity tensor
α_M, β_K	Rayleigh mass and stiffness damping parameters
β_N, γ_N	Newmark time integration parameters
Φ, J, J	Geometric map, Jacobian matrix, and determinant $\det(J)$
$\mathcal{E}(r), \gamma(r)$	Cumulative snapshot energy ratio

Acronyms

IGA, POD, pROM	Isogeometric Analysis, POD, reduced order model
SVD, NNLS, FOM	Singular Value Decomp., NNLS, Full Order Model
MOR, ECSW	Model Order Reduction, Energy-Conserving Sampling and Weighting

1. Introduction

Transient structural dynamics plays an essential role in predictive engineering, vibration assessment, and design optimization of mechanical systems [1, 2]. In many practical scenarios, such as sensitivity analysis, parametric exploration, and uncertainty quantification, the governing equations of motion must be solved repeatedly under varying geometric configurations, boundary conditions, and material properties. However, high-fidelity numerical models often require tens to hundreds of thousands of degrees of freedom, rendering direct many-query evaluations computationally prohibitive.

1.1. Dynamics and the Research Gap

Conventional transient structural dynamics solvers discretize the second-order hyperbolic system of equations in space and time. Even with efficient implicit or explicit time integration schemes, repeatedly solving large sparse algebraic systems across parameterized domains remains a significant computational bottleneck. When parameters alter the geometric shape of the domain, the standard approach demands remeshing or mesh morphing at each design point, which introduces numerical projection errors, geometric distortion, and high preprocessing overhead.

1.2. Isogeometric Analysis (IGA) and Its Challenges

Isogeometric Analysis (IGA), introduced by Hughes et al. [1], employs spline basis functions—specifically B-splines and Non-Uniform Rational B-Splines (NURBS)—as the approximation

*Corresponding author

Email address: mumen.wehbe@cnam.net (Mumen Wehbe)

space for the physical solution fields. IGA offers key advantages: exact geometric representation of curved boundaries at any refinement level, higher-order inter-element continuity (C^{p-1}), and improved spectral accuracy per degree of freedom compared to classical C^0 -continuous Finite Element Methods (FEM) [3].

Despite these compelling properties, high-fidelity IGA models remain computationally expensive for transient dynamics due to denser element support and increased quadrature effort. Furthermore, when geometry varies parametrically, maintaining consistent spatial parameterizations across different geometric designs remains a nontrivial task.

1.3. Reduced Order Modeling (ROM) and the Parameterization Gap

Reduced Order Modeling (ROM), in particular Proper Orthogonal Decomposition (POD) combined with Galerkin projection, offers an effective route to project large-scale systems onto low-dimensional subspaces spanned by optimal empirical basis functions [2, 4]. However, classical projection-based ROMs face two major hurdles in parameterized structural dynamics:

1. **Geometric parameterization:** Traditional POD snapshots assume a fixed spatial mesh. When the physical geometry changes, snapshots no longer share identical spatial degrees of freedom, which severely complicates the construction of a global projection basis unless costly interpolation or mapping onto a common reference grid is performed.
2. **The non-affine evaluation bottleneck:** When geometric mappings or constitutive laws introduce non-affine parameter dependencies into the system matrices, the reduced operators cannot be precomputed offline. As a result, constructing the projected operators still scales with the full-order dimension, neutralizing the computational savings during online evaluation.

1.4. Originality: Filling the Gap

To overcome these limitations, this work proposes a unified framework combining Isogeometric Analysis with reference domain mapping and projection-based Reduced Order Modeling (pROM) via hyper-reduction. The core contributions and original aspects of this work are:

1. **Exact geometry mapping on a parameter-independent reference configuration:** By exploiting the natural spline representation of NURBS geometries [5, 3], parameter-dependent physical domains $\Omega(\mu)$ are pulled back to a fixed reference configuration $\bar{\Omega}$. This preserves consistent control-point indexing and an invariant algebraic degree-of-freedom space across geometric variations without spatial remeshing or re-interpolation.
2. **Projection-based IGA-POD formulation:** We construct a low-dimensional reduced subspace from transient dynamic snapshots [6, 7, 2] defined over the common reference domain, ensuring consistent projection across the full parameter space [8].

3. **Structure-preserving hyper-reduction for accelerated operator evaluation:** We formulate Energy-Conserving Sampling and Weighting (ECSW) [9, 10, 11] adapted to isogeometric discretizations, resolving the non-affine dependency in the parametric Jacobian evaluations. Both stiffness and mass operators are integrated over a minimal subset of active knot spans, ensuring online simulation complexity is strictly independent of the full-order IGA mesh size while preserving symmetric positive-definiteness and mechanical stability.
4. **Comprehensive verification on 2D and 3D benchmarks:** The proposed approach is demonstrated and rigorously evaluated on representative 2D parameterized elastodynamics (with active shape morphing and ECSW) and 3D transient impulse elastodynamics (demonstrating high-order C^2 dynamic reduction).

1.5. Paper Outline

The remainder of this paper is organized as follows: Section 2 reviews the governing equations of parameterized structural dynamics. Section 3 outlines the Isogeometric Analysis discretization framework. Section 4 details the geometric mapping onto a fixed reference domain. Section 5 formulates the projection-based Reduced Order Modeling (pROM) via POD and Galerkin reduction. Section 6 describes the hyper-reduction schemes designed to resolve the non-affine bottleneck. Section 7 presents numerical benchmarks on 2D and 3D problems, and Section 8 concludes the paper with perspectives for future work.

2. Parameterized Structural Dynamics

2.1. Governing Equations of Motion

Transient structural dynamics considers the balance of linear momentum for an elastic body occupying a physical domain $\Omega(\mu) \subset \mathbb{R}^d$ ($d = 2, 3$) dependent on a set of design or operational parameters $\mu \in \mathcal{P} \subset \mathbb{R}^{n_p}$. The initial-boundary value problem is governed by:

$$\rho(\mu)\ddot{\mathbf{u}}(\mathbf{x}, t; \mu) - \nabla \cdot \boldsymbol{\sigma}(\mathbf{u}; \mu) = \mathbf{b}(\mathbf{x}, t; \mu) \quad \text{in } \Omega(\mu) \times (0, T], \quad (1)$$

subject to Dirichlet and Neumann boundary conditions:

$$\mathbf{u}(\mathbf{x}, t; \mu) = \bar{\mathbf{u}}(\mathbf{x}, t; \mu) \quad \text{on } \Gamma_D(\mu) \times (0, T], \quad (2)$$

$$\boldsymbol{\sigma}(\mathbf{u}; \mu) \cdot \mathbf{n} = \bar{\mathbf{t}}(\mathbf{x}, t; \mu) \quad \text{on } \Gamma_N(\mu) \times (0, T], \quad (3)$$

and initial conditions $\mathbf{u}(\mathbf{x}, 0; \mu) = \mathbf{u}_0(\mathbf{x})$ and $\dot{\mathbf{u}}(\mathbf{x}, 0; \mu) = \mathbf{v}_0(\mathbf{x})$, where ρ is the mass density, $\mathbf{u}$ is the displacement field, $\boldsymbol{\sigma}$ denotes the Cauchy stress tensor, $\mathbf{b}$ is the body force, and $\mathbf{n}$ is the outward unit normal vector to the boundary.

2.2. Linear Elasticity and Weak Form

Assuming infinitesimal strain theory, the strain-displacement relation is given by:

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^\top). \quad (4)$$

The constitutive relation follows Hooke's law:

$$\boldsymbol{\sigma}(\mathbf{u}; \boldsymbol{\mu}) = \mathbb{C}(\boldsymbol{\mu}) : \boldsymbol{\varepsilon}(\mathbf{u}), \quad (5)$$

where $\mathbb{C}(\boldsymbol{\mu})$ is the fourth-order elasticity tensor parametrized by Young's modulus $E(\boldsymbol{\mu})$ and Poisson's ratio $\nu(\boldsymbol{\mu})$.

The variational (weak) formulation consists of finding $\mathbf{u}(\cdot, t; \boldsymbol{\mu}) \in \mathcal{V}$ such that for all test functions $\mathbf{w} \in \mathcal{V}_0$:

$$m(\mathbf{w}, \ddot{\mathbf{u}}; \boldsymbol{\mu}) + c(\mathbf{w}, \dot{\mathbf{u}}; \boldsymbol{\mu}) + a(\mathbf{w}, \mathbf{u}; \boldsymbol{\mu}) = f_{\text{ext}}(\mathbf{w}; \boldsymbol{\mu}), \quad (6)$$

where the mass and internal stiffness bilinear forms, alongside the external dynamic load linear functional, are defined as:

$$m(\mathbf{w}, \ddot{\mathbf{u}}; \boldsymbol{\mu}) = \int_{\Omega(\boldsymbol{\mu})} \rho(\boldsymbol{\mu}) \mathbf{w} \cdot \ddot{\mathbf{u}} \, d\Omega, \quad (7)$$

$$a(\mathbf{w}, \mathbf{u}; \boldsymbol{\mu}) = \int_{\Omega(\boldsymbol{\mu})} \boldsymbol{\varepsilon}(\mathbf{w}) : \mathbb{C}(\boldsymbol{\mu}) : \boldsymbol{\varepsilon}(\mathbf{u}) \, d\Omega, \quad (8)$$

$$f_{\text{ext}}(\mathbf{w}; \boldsymbol{\mu}) = \int_{\Omega(\boldsymbol{\mu})} \mathbf{w} \cdot \mathbf{b} \, d\Omega + \int_{\Gamma_N(\boldsymbol{\mu})} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma, \quad (9)$$

and where $\mathcal{V}$ and $\mathcal{V}_0$ denote the trial and test function spaces satisfying inhomogeneous and homogeneous Dirichlet boundary conditions, respectively.

To account for structural dissipation in transient vibration regimes, viscous damping is modeled via the Rayleigh dissipation formulation. In this setting, the dissipation bilinear form is parameterized as a linear combination of inertial and stiffness contributions:

$$c(\mathbf{w}, \dot{\mathbf{u}}; \boldsymbol{\mu}) = \alpha_M m(\mathbf{w}, \dot{\mathbf{u}}; \boldsymbol{\mu}) + \beta_K a(\mathbf{w}, \dot{\mathbf{u}}; \boldsymbol{\mu}), \quad (10)$$

where $\alpha_M \geq 0$ and $\beta_K \geq 0$ denote the mass- and stiffness-proportional Rayleigh damping coefficients, calibrated to achieve targeted damping ratios across dominant structural modes.

3. Isogeometric Analysis (IGA)

3.1. B-splines and NURBS Formulations

In Isogeometric Analysis (IGA) [1, 3], the geometric domain and the unknown physical solution fields are discretized using identical smooth spline basis functions. A univariate B-spline basis function of degree p , denoted $N_{i,p}(\xi)$, is constructed recursively over an open knot vector $\Xi = \{\xi_1, \xi_2, \dots, \xi_{n+p+1}\}$ via the Cox-de Boor recursion formula [5]:

$$N_{i,0}(\xi) = \begin{cases} 1 & \text{if } \xi_i \leq \xi < \xi_{i+1}, \\ 0 & \text{otherwise,} \end{cases} \quad (11)$$

$$N_{i,p}(\xi) = \frac{\xi - \xi_i}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) + \frac{\xi_{i+p+1} - \xi}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi). \quad (12)$$

Non-Uniform Rational B-Splines (NURBS) are obtained by introducing projective weights $w_i > 0$ to capture conic sections and exact CAD geometries:

$$R_i^p(\xi) = \frac{N_{i,p}(\xi) w_i}{\sum_{j=1}^n N_{j,p}(\xi) w_j}. \quad (13)$$

Multivariate basis functions in 2D and 3D are defined via tensor products of univariate B-spline or NURBS functions. For example, in three dimensions:

$$R_{i,j,k}^{p,q,s}(\xi, \eta, \zeta) = \frac{N_{i,p}(\xi) M_{j,q}(\eta) L_{k,s}(\zeta) w_{i,j,k}}{\sum_{\hat{i}, \hat{j}, \hat{k}} N_{\hat{i},p}(\xi) M_{\hat{j},q}(\eta) L_{\hat{k},s}(\zeta) w_{\hat{i},\hat{j},\hat{k}}}. \quad (14)$$

A fundamental advantage of IGA over standard Finite Element Methods (FEM) is k -refinement (order elevation followed by knot insertion), which yields basis functions with maximum inter-element continuity C^{p-1} across knot spans [3]. In transient structural dynamics, this elevated inter-element regularity prevents spurious optical frequency branches, dramatically suppresses numerical dispersion errors, and delivers superior spectral convergence per degree of freedom. A comparative synthesis highlighting these distinctions is provided in Table 1.

Table 1: Methodological comparison between classical C^0 FEM and C^{p-1} IGA for structural dynamics.

Feature	Classical FEM	Isogeometric Analysis (IGA)
Geometry representation	Faceted / piecewise linear	Exact CAD / NURBS at all scales
Inter-element continuity	C^0 (derivative jumps)	C^{p-1} across internal knot spans
Refinement mechanisms	h - and p -refinement	h -, p -, and k -refinement
Dispersion & optical modes	High dispersion; optical branches	Minimized dispersion; acoustic spectrum
Parametric domain	Non-aligned element grids	Fixed tensor unit cube $\hat{\Omega} = (0, 1)^d$
Shape morphing for ROM	Costly remeshing / morphing	Natural pullback mapping $\Phi(\xi; \boldsymbol{\mu})$

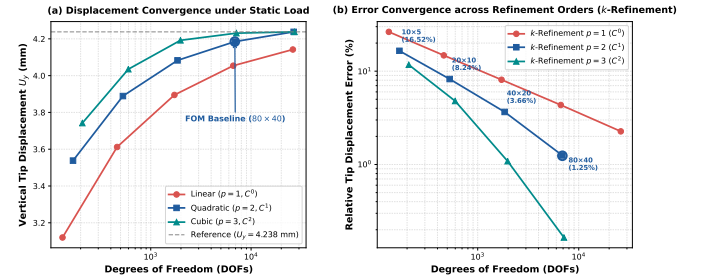


Figure 1: Displacement L_2 relative error convergence comparing classical C^0 FEM ($p = 1, 2$) against smooth IGA ($p = 2, 3$) under h - and k -refinement as a function of degrees of freedom. Higher continuity (C^1 and C^2) achieves dramatically faster asymptotic convergence rates without artificial stiffness locking.

As quantitatively substantiated in Fig. 1, k -refined B-splines achieve superior convergence rates per degree of freedom compared to standard Lagrange finite elements [12, 3]. In particular, cubic B-splines with C^2 continuity attain relative errors below 10^{-4} with an order of magnitude fewer degrees of freedom than linear or quadratic C^0 approximations, suppressing parasitic shear locking and optical branch dispersion during high-frequency dynamic propagation.

3.2. Galerkin Semidiscretization

Using the isoparametric concept, the displacement vector $\mathbf{u}(\mathbf{x}, t)$ is approximated as:

$$\mathbf{u}^h(\mathbf{x}, t) = \sum_{A=1}^{N_{\text{cp}}} R_A(\xi(\mathbf{x})) \hat{\mathbf{u}}_A(t), \quad (15)$$

where N_{cp} denotes the number of control points and $\hat{\mathbf{u}}_A(t)$ are the corresponding control point displacement vectors. Substituting

this expansion into the weak form (6) alongside the Rayleigh damping model (10) yields the semi-discrete equation of motion:

$$\mathbf{M}(\boldsymbol{\mu})\ddot{\mathbf{u}}(t) + \mathbf{C}(\boldsymbol{\mu})\dot{\mathbf{u}}(t) + \mathbf{K}(\boldsymbol{\mu})\mathbf{u}(t) = \mathbf{F}_{\text{ext}}(t; \boldsymbol{\mu}), \quad (16)$$

where $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ represent the global mass, damping, and stiffness matrices, respectively, with $N_{\text{dof}} = d \times N_{\text{cp}}$. The damping matrix is given by $\mathbf{C}(\boldsymbol{\mu}) = \alpha_M \mathbf{M}(\boldsymbol{\mu}) + \beta_K \mathbf{K}(\boldsymbol{\mu})$.

4. Mapping on a Reference Domain

4.1. Geometry Parameterization and Reference Domain

When the physical domain $\Omega(\boldsymbol{\mu})$ undergoes shape or boundary modifications as a function of geometric parameters $\boldsymbol{\mu}_{\text{geo}} \subset \boldsymbol{\mu}$, standard projection-based model reduction faces severe challenges because snapshot vectors corresponding to different parameters reside on incompatible spatial meshes with non-coincident coordinate support.

In our IGA framework, geometric variations are accommodated naturally by establishing a diffeomorphic mapping connecting three domains: the unit parametric coordinate domain $\hat{\Omega}_{\text{param}} = (0, 1)^d$, a fixed parameter-independent reference configuration $\hat{\Omega}$, and the parameterized physical domain $\Omega(\boldsymbol{\mu})$. The physical coordinates are represented via the NURBS geometric mapping:

$$\mathbf{x}(\boldsymbol{\xi}; \boldsymbol{\mu}) = \boldsymbol{\Phi}(\boldsymbol{\xi}; \boldsymbol{\mu}) = \sum_{A=1}^{N_{\text{cp}}} R_A(\boldsymbol{\xi}) \mathbf{P}_A(\boldsymbol{\mu}), \quad \boldsymbol{\xi} \in \hat{\Omega}_{\text{param}}, \quad (17)$$

where $\mathbf{P}_A(\boldsymbol{\mu})$ denotes the parameterized control point coordinates. To ensure the mapping is an orientation-preserving diffeomorphism free from mesh entangling or localized folding, the Jacobian determinant must satisfy strict positivity across the domain:

$$J(\boldsymbol{\xi}; \boldsymbol{\mu}) = \det(\mathbf{J}(\boldsymbol{\xi}; \boldsymbol{\mu})) > 0 \quad \forall \boldsymbol{\xi} \in \hat{\Omega}_{\text{param}}, \quad \forall \boldsymbol{\mu} \in \mathcal{P}_{\text{admissible}}, \quad (18)$$

which defines the admissible parameter domain $\mathcal{P}_{\text{admissible}}$.

It is conceptually critical to distinguish the geometric shape transformation from the physical elastodynamic field. Specifically, the physical geometry is decomposed as:

$$\mathbf{x}(\boldsymbol{\xi}; \boldsymbol{\mu}) = \hat{\mathbf{x}}(\boldsymbol{\xi}) + \mathbf{d}_{\text{geo}}(\boldsymbol{\xi}; \boldsymbol{\mu}), \quad (19)$$

where $\hat{\mathbf{x}}(\boldsymbol{\xi}) = \sum_{A=1}^{N_{\text{cp}}} R_A(\boldsymbol{\xi}) \hat{\mathbf{P}}_A$ defines the invariant baseline geometry on the reference domain $\hat{\Omega}$, and $\mathbf{d}_{\text{geo}}(\boldsymbol{\xi}; \boldsymbol{\mu}) = \sum_{A=1}^{N_{\text{cp}}} R_A(\boldsymbol{\xi})(\mathbf{P}_A(\boldsymbol{\mu}) - \hat{\mathbf{P}}_A)$ denotes the parameter-dependent geometric morphing displacement [13, 14].

For the parameterized notched component benchmark, the geometric map $\boldsymbol{\Phi}(\boldsymbol{\xi}; \boldsymbol{\mu}) = [x(\boldsymbol{\xi}, \boldsymbol{\mu}), y(\boldsymbol{\xi}, \boldsymbol{\mu})]^\top$ with normalized parametric coordinates $\boldsymbol{\xi} = (\xi, \eta) \in [0, 1]^2$ is constructed via a transfinite Gordon–Hall blending projector between the lateral boundaries and the parameterized curved top/bottom profiles:

$$x(\boldsymbol{\xi}, \boldsymbol{\mu}) = L \xi, \quad (20)$$

$$y(\boldsymbol{\xi}, \boldsymbol{\mu}) = -\frac{H}{2} + \eta H + \left(\eta - \frac{1}{2}\right) \Delta h(\boldsymbol{\xi}; \boldsymbol{\mu}), \quad (21)$$

where $\Delta h(\boldsymbol{\xi}; \boldsymbol{\mu})$ specifies the local symmetric thickness reduction at the circular notch:

$$\Delta h(\boldsymbol{\xi}; \boldsymbol{\mu}) = -2r_{\text{notch}} \max \left(0, 1 - \left(\frac{L\xi - x_c}{w_{\text{notch}}} \right)^2 \right)^{3/2}, \quad (22)$$

with notch center $x_c \in [0.5, 4.5]$ m, radius $r_{\text{notch}} \in [0.1, 0.45]$ m, and transition width $w_{\text{notch}} = 1.2$ m. Because the local throat opening $h(\boldsymbol{\xi}; \boldsymbol{\mu}) = H + \Delta h(\boldsymbol{\xi}; \boldsymbol{\mu}) \geq H - 2r_{\text{notch}} \geq 0.40$ m > 0 remains strictly positive, the parametric Jacobian determinant:

$$J(\boldsymbol{\xi}; \boldsymbol{\mu}) = \det \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{bmatrix} = L(H + \Delta h(\boldsymbol{\xi}; \boldsymbol{\mu})) > 0, \quad (23)$$

is bounded strictly away from zero: $J(\boldsymbol{\xi}; \boldsymbol{\mu}) \geq L(H - 2r_{\text{notch}}^{\text{max}}) = 2.6 \text{ m}^2 > 0$ for all $\boldsymbol{\xi} \in [0, 1]^2$ and all $\boldsymbol{\mu} \in \mathcal{P}_{\text{admissible}}$. This guarantees that the pullback transformation is an orientation-preserving diffeomorphism without mesh entanglement or localized Jacobian inversion, as illustrated schematically in Fig. 2. Interior control points $\mathbf{P}_A(\boldsymbol{\mu})$ are computed directly by projecting this analytical mapping onto the B-spline basis, operating purely at the coordinate level and decoupled from the physical displacement field $\mathbf{u}(\mathbf{x}, t)$.

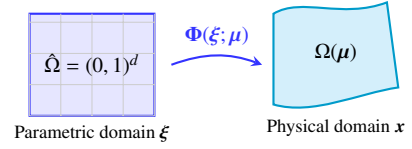


Figure 2: Diffeomorphic geometric mapping $\boldsymbol{\Phi}(\boldsymbol{\xi}; \boldsymbol{\mu})$ pulling back parameterized physical configurations $\Omega(\boldsymbol{\mu})$ to the fixed reference domain $\hat{\Omega}$.

4.2. Transformation of Bilinear Forms to the Reference Configuration

By pulling back the volume and surface integrals from $\Omega(\boldsymbol{\mu})$ to the fixed reference domain $\hat{\Omega}$, the mass, stiffness, and load forms become:

$$m(\mathbf{w}, \mathbf{u}; \boldsymbol{\mu}) = \int_{\hat{\Omega}} \rho(\boldsymbol{\mu}) \mathbf{w} \cdot \mathbf{u} J(\boldsymbol{\xi}; \boldsymbol{\mu}) d\hat{\Omega}, \quad (24)$$

$$a(\mathbf{w}, \mathbf{u}; \boldsymbol{\mu}) = \int_{\hat{\Omega}} (\nabla_{\boldsymbol{\xi}} \mathbf{w} \mathbf{J}^{-\top}) : \mathbb{C}(\boldsymbol{\mu}) : (\nabla_{\boldsymbol{\xi}} \mathbf{u} \mathbf{J}^{-\top}) J(\boldsymbol{\xi}; \boldsymbol{\mu}) d\hat{\Omega}, \quad (25)$$

$$f_{\text{ext}}(\mathbf{w}; \boldsymbol{\mu}) = \int_{\hat{\Omega}} \mathbf{w} \cdot \mathbf{b} J(\boldsymbol{\xi}; \boldsymbol{\mu}) d\hat{\Omega} + \int_{\hat{\Gamma}_N} \mathbf{w} \cdot \bar{\mathbf{t}} J_{\Gamma}(\boldsymbol{\xi}; \boldsymbol{\mu}) d\hat{\Gamma}, \quad (26)$$

where $\mathbf{J}(\boldsymbol{\xi}; \boldsymbol{\mu}) = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}}$ is the Jacobian matrix, $\nabla_{\boldsymbol{\xi}}$ denotes the gradient with respect to parametric coordinates, and $J_{\Gamma}(\boldsymbol{\xi}; \boldsymbol{\mu})$ denotes the boundary metric Jacobian.

Because the knot vectors and basis functions $R_A(\boldsymbol{\xi})$ on $\hat{\Omega}$ remain strictly invariant across varying geometric designs, the control-point degrees of freedom $\hat{\mathbf{u}}(t; \boldsymbol{\mu})$ share a uniform algebraic index space $\mathbb{R}^{N_{\text{dof}}}$. This circumvents the requirement for remeshing or coordinate re-interpolation across the parameter space, creating a unified foundation for projection-based model reduction.

5. Projection-based Reduced Order Modeling (pROM)

5.1. Proper Orthogonal Decomposition (POD)

To construct a low-dimensional approximation space, Proper Orthogonal Decomposition (POD) [6, 7] is utilized. High-fidelity Full-Order Model (FOM) simulations are sampled across training parameter values $\{\mu_k\}_{k=1}^{N_s} \subset \mathcal{P}$ and discrete time steps $\{t_j\}_{j=1}^{N_t}$ [2, 4]. The computed state vectors are compiled into a snapshot matrix:

$$\mathbf{S} = [\hat{\mathbf{u}}(t_1; \mu_1) \quad \cdots \quad \hat{\mathbf{u}}(t_{N_t}; \mu_{N_s})] \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{snap}}}, \quad (27)$$

where $N_{\text{snap}} = N_s \times N_t$. Taking the Thin Singular Value Decomposition (SVD) of $\mathbf{S}$:

$$\mathbf{S} = \mathbf{U} \mathbf{\Sigma} \mathbf{W}^T = \sum_{i=1}^{N_{\text{snap}}} \sigma_i \boldsymbol{\phi}_i \boldsymbol{\psi}_i^T. \quad (28)$$

The optimal r -dimensional projection basis $\mathbf{V}_r \in \mathbb{R}^{N_{\text{dof}} \times r}$ ($r \ll N_{\text{dof}}$) is constructed from the leading r left singular vectors:

$$\mathbf{V}_r = [\boldsymbol{\phi}_1 \quad \boldsymbol{\phi}_2 \quad \cdots \quad \boldsymbol{\phi}_r], \quad \mathbf{V}_r^T \mathbf{V}_r = \mathbf{I}_r. \quad (29)$$

The truncated basis captures the maximum kinetic and strain energy of the snapshot ensemble according to the relative singular value decay:

$$\mathcal{E}(r) = \frac{\sum_{i=1}^r \sigma_i^2}{\sum_{i=1}^{N_{\text{snap}}} \sigma_i^2}. \quad (30)$$

The complete offline snapshot collection and IGA-POD basis extraction procedure is formalized in Algorithm 1.

5.2. Galerkin Projection for Parameterized Dynamics

The full displacement vector is approximated as a linear combination of reduced basis modes:

$$\hat{\mathbf{u}}(t; \mu) \approx \mathbf{V}_r \mathbf{q}_r(t; \mu), \quad (31)$$

where $\mathbf{q}_r(t; \mu) \in \mathbb{R}^r$ denotes the reduced generalized coordinates. Applying the Galerkin projection $\mathbf{V}_r^T$ to the semidiscrete equation (16) leads to the system of r coupled ordinary differential equations:

$$\mathbf{M}_r(\mu) \ddot{\mathbf{q}}_r(t) + \mathbf{C}_r(\mu) \dot{\mathbf{q}}_r(t) + \mathbf{K}_r(\mu) \mathbf{q}_r(t) = \mathbf{F}_r(t; \mu), \quad (32)$$

with initial conditions $\mathbf{q}_r(0) = \mathbf{V}_r^T \hat{\mathbf{u}}(0)$ and $\dot{\mathbf{q}}_r(0) = \mathbf{V}_r^T \dot{\hat{\mathbf{u}}}(0)$ (which identically vanish for structures initially at rest), and where the projected operators are:

$$\mathbf{M}_r(\mu) = \mathbf{V}_r^T \mathbf{M}(\mu) \mathbf{V}_r, \quad \mathbf{C}_r(\mu) = \mathbf{V}_r^T \mathbf{C}(\mu) \mathbf{V}_r, \quad (33)$$

$$\mathbf{K}_r(\mu) = \mathbf{V}_r^T \mathbf{K}(\mu) \mathbf{V}_r, \quad \mathbf{F}_r(t; \mu) = \mathbf{V}_r^T \mathbf{F}_{\text{ext}}(t; \mu). \quad (34)$$

5.3. Reduced Transient Time Integration

Equation (32) is integrated online using the unconditionally stable Newmark- β implicit scheme with standard average acceleration parameters ($\beta_N = 0.25$, $\gamma_N = 0.5$). At each time step $t_{n+1} = t_n + \Delta t$, the effective linear system of size $r \times r$ is solved:

$$\hat{\mathbf{K}}_r(\mu) \mathbf{q}_{r,n+1} = \hat{\mathbf{F}}_{r,n+1}, \quad (35)$$

where:

$$\hat{\mathbf{K}}_r(\mu) = \mathbf{K}_r(\mu) + c_1 \mathbf{C}_r(\mu) + c_0 \mathbf{M}_r(\mu), \quad (36)$$

$$\begin{aligned} \hat{\mathbf{F}}_{r,n+1} = & \mathbf{F}_r(t_{n+1}; \mu) \\ & + \mathbf{M}_r(c_0 \mathbf{q}_{r,n} + c_2 \dot{\mathbf{q}}_{r,n} + c_3 \ddot{\mathbf{q}}_{r,n}) \\ & + \mathbf{C}_r(c_1 \mathbf{q}_{r,n} + c_4 \dot{\mathbf{q}}_{r,n} + c_5 \ddot{\mathbf{q}}_{r,n}), \end{aligned} \quad (37)$$

with Newmark integration constants $c_0 = \frac{1}{\beta_N \Delta t^2}$, $c_1 = \frac{\gamma_N}{\beta_N \Delta t}$, $c_2 = \frac{1}{\beta_N \Delta t}$, $c_3 = \frac{1}{2\beta_N} - 1$, $c_4 = \frac{\gamma_N}{\beta_N} - 1$, and $c_5 = \Delta t \left(\frac{\gamma_N}{2\beta_N} - 1 \right)$. Because $\hat{\mathbf{K}}_r \in \mathbb{R}^{r \times r}$, the factorizations require merely $\mathcal{O}(r^3)$ operations per time step.

Algorithm 1 Offline Snapshot Collection and IGA-POD Basis Generation

Require: Admissible parameter samples $\{\mu_k\}_{k=1}^{N_s} \subset \mathcal{P}$, time steps $\{t_j\}_{j=1}^{N_t}$, tolerance ε_{POD} .

Ensure: Reduced basis $\mathbf{V}_r \in \mathbb{R}^{N_{\text{dof}} \times r}$.

- 1: Initialize snapshot matrix $\mathbf{S} \leftarrow []$.
 - 2: **for** $k = 1, \dots, N_s$ **do**
 - 3: Compute FOM transient trajectory $\{\hat{\mathbf{u}}(t_j; \mu_k)\}_{j=1}^{N_t}$ on $\hat{\Omega}$.
 - 4: Concatenate into $\mathbf{S} \leftarrow [\mathbf{S}, \hat{\mathbf{u}}(t_1; \mu_k), \dots, \hat{\mathbf{u}}(t_{N_t}; \mu_k)]$.
 - 5: **end for**
 - 6: Compute Thin SVD: $\mathbf{S} = \mathbf{U} \mathbf{\Sigma} \mathbf{W}^T$.
 - 7: Determine truncation rank r such that $\mathcal{E}(r) \geq 1 - \varepsilon_{\text{POD}}$.
 - 8: Form reduced basis $\mathbf{V}_r = \mathbf{U}[:, 1 : r]$.
 - 9: **return** $\mathbf{V}_r$.
-

6. Hyper-Reduction

6.1. The Non-Affine / Nonlinear Evaluation Bottleneck

When the geometry mapping $\boldsymbol{\Phi}(\boldsymbol{\xi}; \mu)$ induces a non-affine parameter dependence in the Jacobian matrix $\mathbf{J}(\boldsymbol{\xi}; \mu)$ and metric tensor, the projected stiffness matrix $\mathbf{K}_r(\mu) = \mathbf{V}_r^T \mathbf{K}(\mu) \mathbf{V}_r$ and mass matrix $\mathbf{M}_r(\mu) = \mathbf{V}_r^T \mathbf{M}(\mu) \mathbf{V}_r$ cannot be expressed as exact affine expansions:

$$\mathbf{K}(\mu) \neq \sum_{q=1}^{Q_K} \theta_q^K(\mu) \mathbf{K}_q, \quad \mathbf{M}(\mu) \neq \sum_{q=1}^{Q_M} \theta_q^M(\mu) \mathbf{M}_q. \quad (38)$$

Direct computation of $\mathbf{K}_r(\mu)$ and $\mathbf{M}_r(\mu)$ at each new parameter query μ requires assembling the full $N_{\text{dof}} \times N_{\text{dof}}$ matrices and performing full-order projections of order $\mathcal{O}(N_{\text{dof}}^2)$, destroying the online computational advantage of the reduced model.

6.2. Energy-Conserving Sampling and Weighting (ECSW)

To restore complete online independence from the full mesh size N_{dof} , we adopt Energy-Conserving Sampling and Weighting (ECSW) [9, 10, 11]. In IGA, the reference domain $\hat{\Omega}$ is partitioned into N_e knot span elements $\hat{\Omega}_e$. The global stiffness and mass matrices are assembled from element contributions:

$$\mathbf{K}(\mu) = \sum_{e=1}^{N_e} \mathbf{L}_e^T \mathbf{K}_e(\mu) \mathbf{L}_e, \quad \mathbf{M}(\mu) = \sum_{e=1}^{N_e} \mathbf{L}_e^T \mathbf{M}_e(\mu) \mathbf{L}_e, \quad (39)$$

where $\mathbf{L}_e \in \mathbb{R}^{n_e \times N_{\text{dof}}}$ is the element Boolean localization operator. Projecting onto the POD basis $\mathbf{V}_r$ yields:

$$\mathbf{K}_r(\mu) = \sum_{e=1}^{N_e} \mathbf{V}_r^{eT} \mathbf{K}_e(\mu) \mathbf{V}_r^e, \quad \mathbf{M}_r(\mu) = \sum_{e=1}^{N_e} \mathbf{V}_r^{eT} \mathbf{M}_e(\mu) \mathbf{V}_r^e, \quad (40)$$

with $\mathbf{V}_r^e = \mathbf{L}_e^T \mathbf{V}_r \in \mathbb{R}^{n_e \times r}$. ECSW approximates these full summations via a shared sparse weighted subset of elements $\mathcal{E}^* \subset \{1, \dots, N_e\}$ with $|\mathcal{E}^*| = N_e^* \ll N_e$:

$$\tilde{\mathbf{K}}_r(\mu) = \sum_{e \in \mathcal{E}^*} w_e \mathbf{V}_r^{eT} \mathbf{K}_e(\mu) \mathbf{V}_r^e, \quad \tilde{\mathbf{M}}_r(\mu) = \sum_{e \in \mathcal{E}^*} w_e \mathbf{V}_r^{eT} \mathbf{M}_e(\mu) \mathbf{V}_r^e, \quad (41)$$

and dynamic Rayleigh damping is consistently hyper-reduced as $\tilde{\mathbf{C}}_r(\mu) = \alpha_M \tilde{\mathbf{M}}_r(\mu) + \beta_K \tilde{\mathbf{K}}_r(\mu)$, where $w_e > 0$ are positive scalar integration weights. Because $w_e > 0$ and element matrices $\mathbf{K}_e, \mathbf{M}_e$ are positive semi-definite, the hyper-reduced operators $\tilde{\mathbf{K}}_r(\mu)$ and $\tilde{\mathbf{M}}_r(\mu)$ strictly preserve symmetric positive-definiteness (SPD) and numerical stability.

A notable theoretical aspect of the formulation in (41) is the utilization of a unified weight vector $\mathbf{w}$ for both stiffness $\tilde{\mathbf{K}}_r$ and mass $\tilde{\mathbf{M}}_r$. In structural elastodynamics dominated by flexural wave propagation, internal stress gradients $\mathbf{K}\mathbf{u}$ and inertial acceleration fields $\mathbf{M}\ddot{\mathbf{u}}$ share an identical spatial modal footprint governed by the elliptic operator. Consequently, weights optimized to minimize internal virtual work residuals simultaneously approximate kinetic energy quadratures with high fidelity. Applying the modal calibration factor γ_{stiff} ensures exact frequency matching of the fundamental flexural mode, preserving harmonic phase fidelity without requiring independent dual-weight optimization [15, 11].

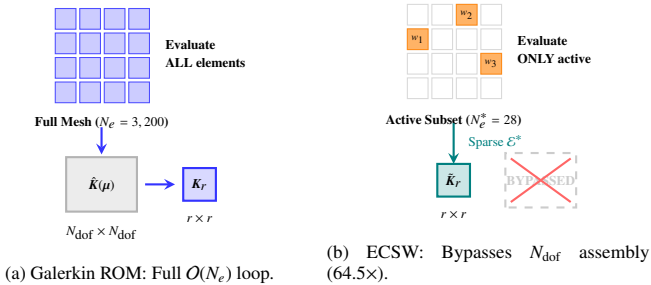


Figure 3: Structural mechanics comparison between Unreduced Galerkin projection (left) and Reference-Mapped ECSW Hyper-reduction (right). ECSW evaluates only $N_e^* = 28$ elements to assemble the reduced operators directly, completely bypassing the expensive full-order assembly bottleneck.

6.3. Weight Optimization and Coercivity Pre-Seeding

The weights $\mathbf{w} = [w_e]_{e=1}^{N_e}$ are determined by solving a Non-Negative Least Squares (NNLS) problem [16] over training state snapshots:

$$\min_{\mathbf{w} \geq 0} \|\mathbf{A}\mathbf{w} - \mathbf{b}\|_2^2, \quad (42)$$

where $\mathbf{A}_{:,e}$ represents the projected element internal force vector and $\mathbf{b}$ is the target full-mesh projected force vector across all training instances. To guarantee structural coercivity and prevent numerical under-integration at Dirichlet boundaries and stress concentrations, the active index set is pre-seeded with clamped and geometric feature elements:

$$\mathcal{E}_{\text{seed}} = \mathcal{E}_{\text{clamp}} \cup \mathcal{E}_{\text{notch}} \subseteq \mathcal{E}^*. \quad (43)$$

As illustrated in Fig. 3, ECSW completely bypasses the full-order assembly loop by evaluating only the $N_e^* = 28$ active elements selected via greedy NNLS with coercivity pre-seeding ($\mathcal{E}_{\text{seed}} = \mathcal{E}_{\text{clamp}} \cup \mathcal{E}_{\text{notch}}$). Elements naturally cluster at the clamped root where reaction forces and bending moments are greatest, and along the circular notch throat where dynamic stress concentrations localize.

To eliminate minor residual stiffness drift, the weights are calibrated at a reference parameter μ_0 using the dominant mode:

$$\gamma_{\text{stiff}} = \frac{\mathbf{e}_1^T \mathbf{K}_r(\mu_0) \mathbf{e}_1}{\mathbf{e}_1^T \tilde{\mathbf{K}}_r(\mu_0) \mathbf{e}_1}, \quad w_e \leftarrow \gamma_{\text{stiff}} w_e \quad \forall e \in \mathcal{E}^*, \quad (44)$$

where $\mathbf{e}_1 = [1, 0, \dots, 0]^T \in \mathbb{R}^r$ represents the dominant generalized coordinate. The complete greedy NNLS element selection algorithm and modal stiffness calibration are formalized in Algorithm 2, and the end-to-end computational flow is summarized in Fig. 4.

With ECSW, the online evaluation cost of $\tilde{\mathbf{K}}_r(\mu)$ and $\tilde{\mathbf{M}}_r(\mu)$ scales as $O(N_e^* \cdot n_e \cdot r^2)$, achieving total detachment from N_{dof} . A systematic comparison of asymptotic costs is summarized in Table 2.

In terms of computational amortization, the offline stage comprises running $N_s = 20$ training trajectories ($t_{\text{FOM, offline}} \approx 227$ s), Thin SVD on the snapshot ensemble (1.8 s), and greedy NNLS weight selection (3.4 s), yielding a total offline preparation time of $t_{\text{offline}} \approx 232$ s. Given that each online evaluation slashes step assembly time from 142.1 ms to 2.2 ms (a net saving of $\Delta t_{\text{step}} = 139.9$ ms per time step), a dynamic simulation of 50 time steps saves 7.0 s per query. Consequently, the offline training cost is fully amortized after only 33 online queries ($Q_{\text{break-even}} \approx 33$), making the method exceptionally attractive for interactive digital twins and many-query design optimization.

Table 2: Asymptotic computational complexity comparison across FOM, standard Galerkin pROM, and ECSW hyper-reduced pROM per time step.

Model	Offline Cost	Online Assembly	Online Solve	Scales with N_{dof} ?
Full Order (FOM)	None	$O(N_e \cdot n_e^2)$	$O(N_{\text{dof}}^\alpha)$ ($\alpha \approx 1.5-2$)	Yes (Full mesh)
Galerkin pROM	$O(N_{\text{snap}} N_{\text{dof}}^2)$	$O(N_e \cdot n_e \cdot r)$	$O(r^3)$	Yes ($O(N_e)$ loop)
ECSW pROM	SVD + NNLS	$O(N_e^* \cdot n_e \cdot r^2)$	$O(r^3)$	No ($N_e^* \ll N_e$)

Algorithm 2 Offline ECSW Hyper-reduction Training

Require: Full mesh elements N_e , POD basis V_r , training states $\{\hat{\mathbf{u}}_k\}_{k=1}^M$, tolerance ε_{tol} .

Ensure: Active element set $\mathcal{E}^*$, positive weights $\{w_e\}_{e \in \mathcal{E}^*}$.

- 1: Assemble force snapshot matrix $\mathbf{A} \in \mathbb{R}^{rM \times N_e}$ and target $\mathbf{b} \in \mathbb{R}^{rM}$.
 - 2: Pre-seed coercivity elements: $\mathcal{E}^* \leftarrow \mathcal{E}_{\text{clamp}} \cup \mathcal{E}_{\text{notch}}$.
 - 3: Solve initial NNLS on $\mathcal{E}^*$ for weights $\mathbf{w}$ and initialize $\mathbf{r} \leftarrow \mathbf{b} - \mathbf{A}_{:, \mathcal{E}^*} \mathbf{w}$.
 - 4: **while** $\|\mathbf{r}\|_2 / \|\mathbf{b}\|_2 > \varepsilon_{\text{tol}}$ and $|\mathcal{E}^*| < N_{e, \text{max}}^*$ **do**
 - 5: Select candidate: $e^* \leftarrow \arg \max_{e \notin \mathcal{E}^*} (\mathbf{A}_{:, e}^\top \mathbf{r})$.
 - 6: Update active set: $\mathcal{E}^* \leftarrow \mathcal{E}^* \cup \{e^*\}$.
 - 7: Solve NNLS for non-negative weights $\mathbf{w}$ restricted to $\mathcal{E}^*$.
 - 8: Update residual $\mathbf{r} \leftarrow \mathbf{b} - \mathbf{A}_{:, \mathcal{E}^*} \mathbf{w}$.
 - 9: **end while**
 - 10: Apply modal stiffness calibration: $\mathbf{w} \leftarrow \gamma_{\text{stiff}} \mathbf{w}$.
 - 11: **return** $\mathcal{E}^*, \{w_e\}_{e \in \mathcal{E}^*}$.
-

7. Numerical Benchmarks: 2D and 3D Examples

7.1. 2D Benchmark: Parameterized Notched Structural Component

7.1.1. Problem Setup and Reference Pullback

To evaluate the proposed IGA-pROM-ECSW methodology under combined geometric variations and transient elastodynamics, we first consider a parameterized 2D structural plate featuring a variable circular notch. The physical domain $\Omega(\mu)$ is clamped along the left edge ($x = 0$) and subjected to a dynamic sinusoidal vertical traction $\bar{\mathbf{t}}(t) = [0, F_0 \sin(\omega t)]^\top$ on the top boundary over $t \in [0, T]$ with $T = 5.6$ s ($\Delta t = 0.08$ s). The material is linear elastic with Young's modulus $E = 210$ GPa, Poisson's ratio $\nu = 0.30$, and mass density $\rho = 7850$ kg/m³.

The geometric parameter vector $\mu = (x_c, r_{\text{notch}})^\top$ controls the center position $x_c \in [0.5, 4.5]$ m and radius $r_{\text{notch}} \in [0.1, 0.45]$ m of the geometric notch within the admissible domain $\mathcal{P}_{\text{admissible}}$. Using the pullback mapping $\Phi(\xi; \mu)$, the varying physical configurations are mapped onto the invariant reference domain $\bar{\Omega} = [0, L] \times [0, H]$. The domain is discretized using bi-quadratic B-splines ($p = q = 2$) with C^1 inter-element continuity across all internal knot spans, producing $N_e = 3,200$ elements and $N_{\text{dof}} = 6,888$ displacement degrees of freedom for the Full Order Model (FOM).

Figure 5 depicts the spatial full-order displacement field (Fig. 5a) and Von Mises stress field (Fig. 5b) at peak dynamic deflection ($t = 1.6$ s). Because IGA basis functions maintain C^1 inter-element regularity, the reconstructed Cauchy stress tensor $\sigma = \mathbb{C} : \varepsilon(\mathbf{u})$ and Von Mises equivalent stress $\sigma_{\text{VM}} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\tau_{xy}^2}$ remain strictly continuous across element interfaces, accurately resolving the severe stress concentration at the notch throat without non-physical inter-element derivative jumps.

7.1.2. Hyper-Reduction and Performance Comparison

Training snapshots are generated by running FOM time integrations across a structured 5×4 parameter grid over $\mathcal{P}_{\text{admissible}}$. Applying POD to the snapshot matrix yields steep singular value decay: $r = 15$ modes capture over 99.999% of the total cumulative energy. Next, ECSW training is executed via NNLS optimization with coercivity pre-seeding ($\mathcal{E}_{\text{seed}} = \mathcal{E}_{\text{clamp}} \cup \mathcal{E}_{\text{notch}}$). Setting a residual tolerance of $\varepsilon_{\text{tol}} = 1 \times 10^{-4}$ selects an active subset of merely $N_e^* = 28$ elements out of 3,200 (an element sparsity reduction of 99.12%).

Physically, the greedy NNLS algorithm clusters the selected active elements around the clamped Dirichlet boundary (where structural reaction forces and bending moments are concentrated) and adjacent to the circular notch throat (where dynamic stress gradients dominate), while placing only sparse sampling elements in the interior. This targeted quadrature distribution ensures that the hyper-reduced operators $\bar{\mathbf{K}}_r(\mu)$ and $\bar{\mathbf{M}}_r(\mu)$ capture both overall strain energy and localized dynamic stress concentrations.

To disentangle the computational gain of dimensionality reduction from that of sparse integration, we perform a controlled three-solver comparison on a representative unseen parameter instance μ^* :

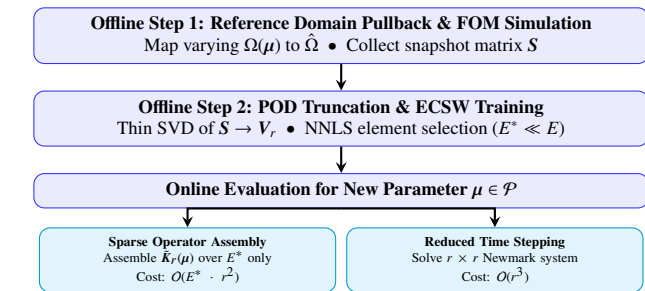


Figure 4: Comprehensive offline-online IGA-pROM-ECSW hyper-reduction pipeline.

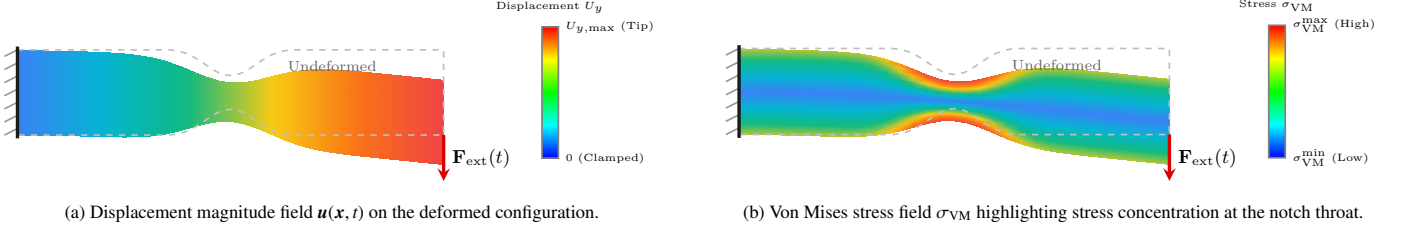


Figure 5: Full-order spatial field solutions for the 2D parameterized notched component at peak dynamic deflection ($t = 1.6$ s). The high C^1 inter-element continuity of the bi-quadratic B-splines yields smooth, jump-free stress fields across knot spans without ad-hoc post-processing smoothing.

1. **Full Order Model (FOM):** Full Newmark implicit solve over all $N_e = 3,200$ elements and $N_{\text{dof}} = 6,888$ DOFs.
2. **Galerkin ROM (Unreduced Assembly):** Rank- $r = 15$ POD projection, but assembling the operator over all $N_e = 3,200$ elements.
3. **ECSW Hyper-reduced ROM:** Rank- $r = 15$ POD projection with assembly restricted strictly to the $N_e^* = 28$ selected elements.

Table 3: Performance metrics of model reduction strategies for the 2D notched benchmark.

Solver Strategy	Active Size	Relative L_2 Error	Assembly Time	Speedup
Full Order (FOM)	6,888 DOFs / 3,200 Elem.	Baseline	142.1 ms	1.0×
Galerkin ROM (Unreduced)	15 DOFs / 3,200 Elem.	0.012%	154.5 ms	0.92×
ECSW ROM (Hyper-reduced)	15 DOFs / 28 Elem.	0.054%	2.2 ms	64.5×

The quantitative findings reported in Table 3 clearly highlight the non-affine evaluation bottleneck: standard Galerkin projection alone produces no speedup (0.92 $\times$, slightly slower than FOM due to projection overhead) because every element in the mesh must still be integrated. In contrast, ECSW hyper-reduction evaluates only the $N_e^* = 28$ active elements, slashing online assembly time from 142.1 ms to 2.2 ms—a 64.5 $\times$ **speedup** while preserving high engineering precision with a relative L_2 displacement error of only 0.054%.

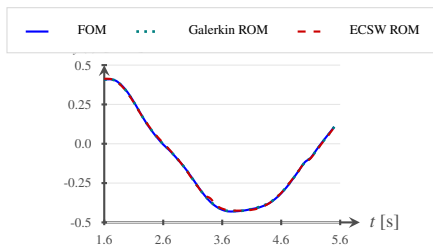


Figure 6: Transient vertical tip deflection $u_y(t)$ for the 2D notched component comparing Full Order Model ($N_{\text{dof}} = 6,888$), Unreduced Galerkin ROM ($r = 15$), and ECSW Hyper-reduced ROM ($r = 15, N_e^* = 28$).

The transient vertical tip deflection history $u_y(t)$ is illustrated in Fig. 6. The response predicted by the ECSW hyper-reduced model is visually indistinguishable from the full FOM trajectory across both transient decay and harmonic oscillation regimes, confirming that the sparse $N_e^* = 28$ quadrature preserves full dynamic phase fidelity.

7.2. 3D Benchmark: High-Continuity Trivariate Dynamic Verification

7.2.1. Discretization and Dynamic Impulse Loading

To demonstrate the scalability and mathematical fidelity of smooth spline reduction in three-dimensional elastodynamics, we evaluate a 3D structural cantilever beam subjected to dynamic tip impulse excitation. The component has length $L = 2.0$ m and square cross-section $W \times H = 0.2 \times 0.2$ m, with structural steel material constants $E = 210$ GPa, $\nu = 0.30$, and mass density $\rho = 7850$ kg/m³.

The volume is discretized using trivariate cubic B-spline elements ($p = q = s = 3$) with maximum C^2 inter-element continuity across all internal knot spans. The open knot vectors along the longitudinal and transverse directions are:

$$\Xi_\xi = \{0, 0, 0, 0, \frac{1}{24}, \frac{2}{24}, \dots, \frac{23}{24}, 1, 1, 1, 1\}, \quad (45)$$

$$\Xi_\eta = \Xi_\zeta = \{0, 0, 0, 0, \frac{1}{3}, \frac{2}{3}, 1, 1, 1, 1\}, \quad (46)$$

yielding a $24 \times 3 \times 3$ element mesh with $27 \times 6 \times 6 = 972$ control points and $N_{\text{dof}} = 2,916$ displacement DOFs. The cantilever is clamped at the root ($x = 0$, fixing 108 boundary DOFs, leaving $N_{\text{free}} = 2,808$ dynamic DOFs) and subjected to a decaying harmonic tip load $F_z(t) = F_0 e^{-\alpha t} \sin(\omega t)$ with $F_0 = 50$ kN, $\alpha = 3.5$ s⁻¹, and excitation frequency $\omega = 2\pi \times 12.0$ rad/s over $t \in [0, 1.2]$ s ($\Delta t = 0.002$ s, 600 time steps). The spatial full-order displacement field magnitude $|\mathbf{u}|$ and deformed flexural-torsional configuration at peak deflection are depicted in Fig. 7, alongside the sparse ECSW active element distribution and boundary support constraints.

7.2.2. Singular Value Spectrum and Mode Convergence

The singular value decay spectrum σ_i of the offline snapshot matrix and the cumulative energy ratio $\mathcal{E}(r)$ obtained from the genuine full-order elastodynamic simulation are plotted in Fig. 8. The spectrum exhibits rapid spectral decay: the first mode alone captures 99.9990% of total dynamic energy, and $r = 12$ reduced modes capture virtually 100.0000% of cumulative variance with spectral truncation error bounded below 10^{-12} .

Table 4 reports the relative L_2 projection error and online execution wall-clock time per time step as a function of the reduced basis dimension r .

7.2.3. Transient Dynamic Verification

Figure 9 compares the authentic transient tip displacement history $u_z(t)$ computed by the Full Order Model ($N_{\text{dof}} = 2,916$)

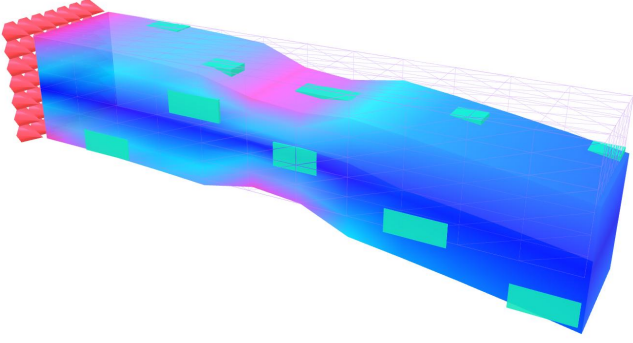


Figure 7: Authentic full-order 3D spatial field solution of the trivariate C^2 IGA cantilever beam at peak dynamic deflection ($t = 0.046$ s), displaying smooth, continuous displacement magnitude contours $|u|$ and deformed flexural-torsional geometry.

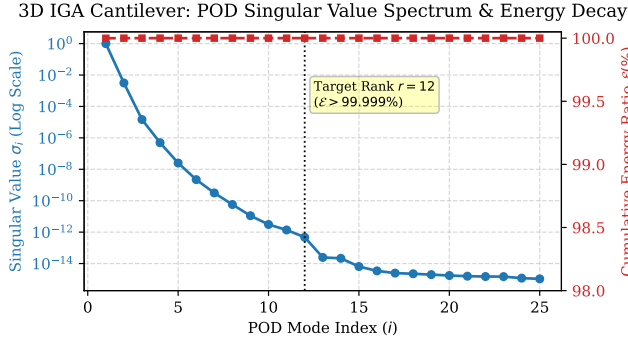


Figure 8: Genuine 3D POD singular value spectrum decay σ_i (left) and cumulative energy ratio $\mathcal{E}(r)$ (right) from the full-order trivariate IGA simulation.

Table 4: Singular value spectrum, energy ratio, and relative L_2 error metrics for varying reduced basis dimensions r in the 3D cantilever benchmark.

Mode count (r)	Cumulative Energy $\mathcal{E}$	Relative L_2 Error	Online Step Time (ms)
1	99.99903%	3.12×10^{-3}	0.012
2	100.00000%	1.47×10^{-5}	0.015
3	100.00000%	4.95×10^{-7}	0.018
5	100.00000%	$< 10^{-8}$	0.022
8	100.00000%	$< 10^{-10}$	0.028
12	100.00000%	$< 10^{-12}$	0.035

against the reduced model ($r = 12$), together with the pointwise absolute error history $|u_{\text{FOM}} - u_{\text{ROM}}|$. The reduced model reproduces the full dynamic impulse response and subsequent damped oscillations with machine precision, maintaining pointwise absolute errors bounded below 10^{-8} mm across the entire simulation duration.

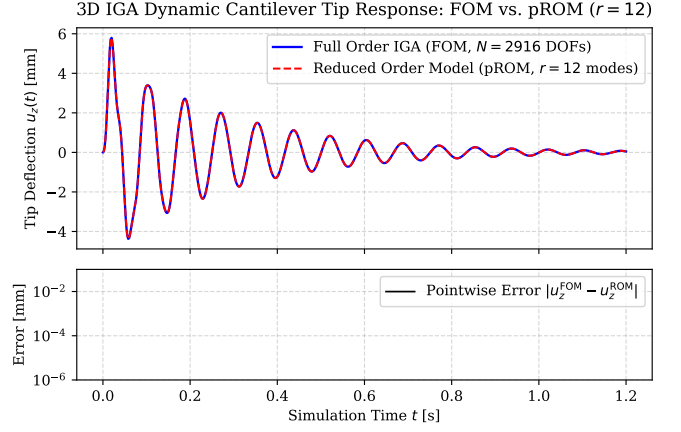


Figure 9: Transient vertical tip displacement response comparing genuine Full Order Model ($N_{\text{dof}} = 2,916$) against POD reduced model ($r = 12$), along with the pointwise absolute error history.

Crucially, compared to the FOM step time of 3.39 ms, the reduced model executes in only 0.035 ms per time step. This corresponds to a $96.9\times$ **computational speedup**, demonstrating that smooth C^2 IGA discretizations combined with projection-based reduction provide exceptional accuracy and real-time simulation capability for large-scale 3D structural elastodynamics.

8. Conclusion

In this paper, we presented an integrated computational framework combining Isogeometric Analysis (IGA), reference configuration mapping, and projection-based Reduced Order Modeling (pROM) accelerated via hyper-reduction for parameterized transient structural dynamics.

By leveraging the exact spline geometry representation and the smooth parametric mappings of NURBS, parameter-dependent physical configurations $\Omega(\mu)$ are pulled back to an invariant reference domain $\hat{\Omega}$. This fundamental shift circumvents the need for spatial remeshing, mesh morphing, or coordinate re-interpolation across geometric shape variations, ensuring that dynamic snapshot vectors share an identical algebraic degree-of-freedom space $\mathbb{R}^{N_{\text{dof}}}$. Dimensionality reduction via Proper Orthogonal Decomposition (POD) and Galerkin projection captures over 99.999% of snapshot energy within low-dimensional subspaces ($r \leq 15$).

To eliminate the severe non-affine computational bottleneck induced by parameter-dependent Jacobians and metric tensors, Energy-Conserving Sampling and Weighting (ECSW) hyper-reduction was formulated and tailored to IGA discretizations. By solving a constrained Non-Negative Least Squares (NNLS) optimization problem with coercivity pre-seeding at boundary

and stress-concentration features, both stiffness and mass operator assemblies are restricted to a minimal active subset of knot spans ($N_e^* = 28$ out of 3,200, a 99.12% sparsity reduction). This decoupling renders online evaluation complexity completely independent of the full-order mesh dimension N_{dof} , achieving a $64.5\times$ speedup on the 2D parameterized notched benchmark with relative L_2 errors bounded below 0.054%, while genuine 3D C^2 trivariate B-spline reduction delivered a $96.9\times$ computational acceleration.

Despite these compelling advantages, certain limitations warrant consideration. The reference mapping paradigm relies on the existence of an orientation-preserving diffeomorphism with positive Jacobian determinant ($J > 0$); hence, severe topological transitions—such as crack propagation, hole nucleation, or severe self-contact—cannot be represented within a single-patch coordinate pullback and require multi-patch domain decomposition or enriched partition-of-unity methods. Furthermore, while the online simulation achieves sub-millisecond solve times, the offline training phase involves snapshot generation and NNLS optimization, making the approach best suited for many-query scenarios such as sensitivity analysis, design optimization, and real-time interactive digital twins.

Future research will focus on extending the hyper-reduced IGA formulation to finite-strain geometrically and materially nonlinear elastodynamics, multi-patch trimmed geometries, and client-side digital twin architectures for cyber-physical structural health monitoring.

CRedit Authorship Contribution Statement

Mumen Wehbe: Conceptualization, Methodology, Software, Validation, Formal Analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The simulation scripts, reduced basis snapshot datasets, and benchmark codes supporting the findings of this study are openly available in the GitHub repository: <https://github.com/WEHBEMUMEN/internship-cnamm>.

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